Lecture 16: Maps continued

Heather Mattie, PhD

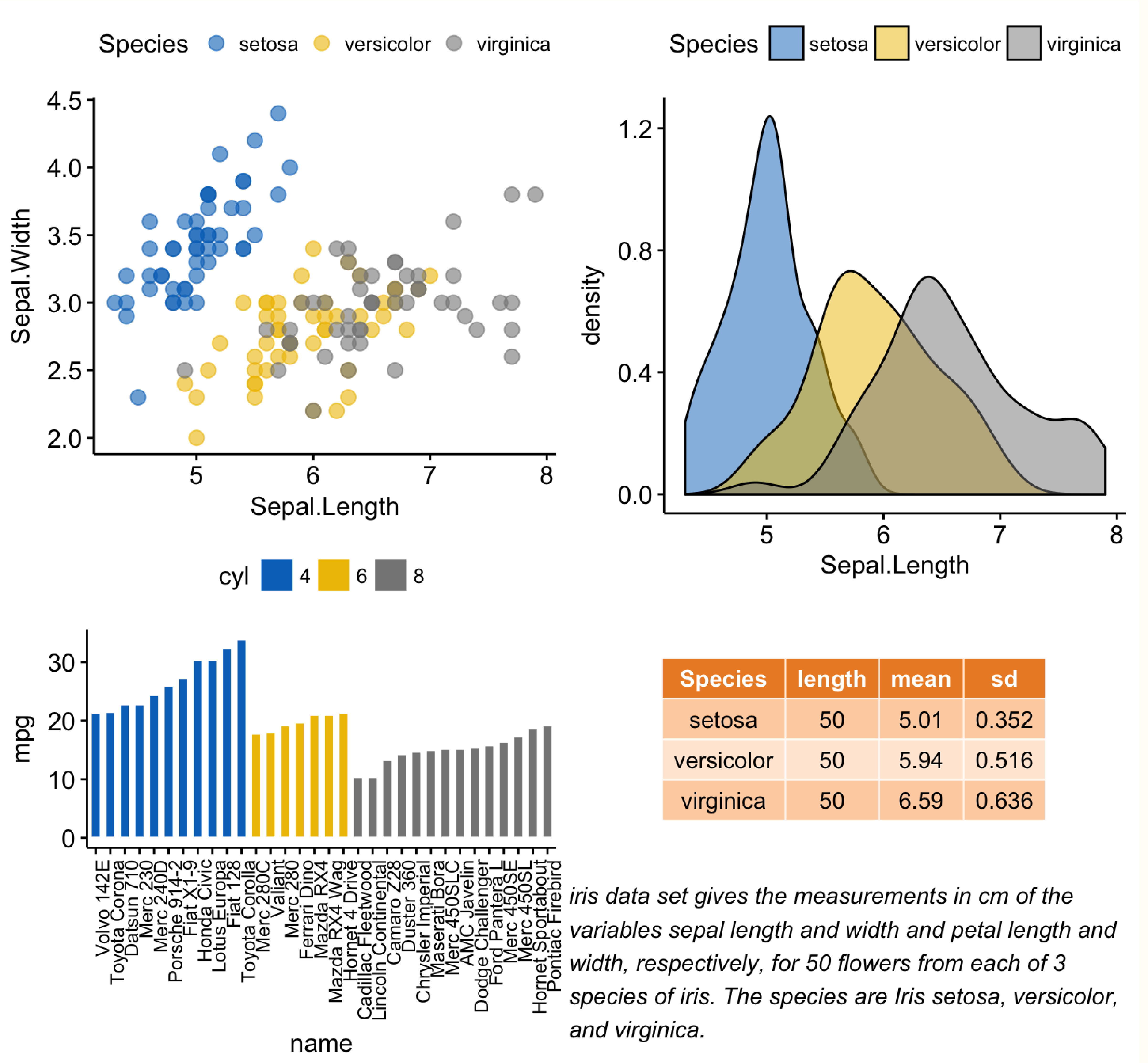
October 23, 2025

Recipe of the Day!

[Harry Potter Sorting Hat Cookies]

Announcements

- Homework #3 due Friday, October 24 by 11:59pm
- No lab this week
- Midterm will be released on **Monday, October 27** and due **Friday, November 7**
    - No late days may be used



Combining plots and tables

Coding question of the day

Using the **`gapminder`** dataset, create a categorical life expectancy variable and add it to the **`gapminder`** dataset. Call this variable **`life_expectancy_category`** and use the **`case_when`** function to create it. Life expectancy below 50 should be labeled "Low", life expectancy greater than or equal to 50 and less than 75 should be labeled "Medium", and life expectancy greater than or equal to 75 should be labeled "High".

Now, filter the dataset to only include observations from the year 1999, and create a bar chart of the 3 life expectancy categories. Reorder the bars so that they are in the order Low, Medium, High.

Be sure to run this code first

```
library(dslabs)
library(dplyr)
library(ggplot2)
data(gapminder)
```

Maps

Plotting US state murder rates

Heather Mattie, PhD

Plotting state murder rates

- Task: plot a map of the US states and fill in each state with a color representing the murder rate in that state using the `murders` dataset
- Need to use state map data and add the murder rate for each state to the map data frame

```r
library(tidyverse)
library(maps)
library(dslabs)
library(RColorBrewer)

data(murders)
murders <- murders %>% mutate(rate = total / population * 100000)
states_map <- map_data("state")

head(murders)
```

```
##        state abb region population total     rate
## 1    Alabama  AL  South    4779736   135 2.824424
## 2     Alaska  AK   West     710231    19 2.675186
## 3    Arizona  AZ   West    6392017   232 3.629527
## 4   Arkansas  AR  South    2915918    93 3.189390
## 5 California  CA   West   37253956  1257 3.374138
## 6   Colorado  CO   West    5029196    65 1.292453
```

```r
head(states_map)
```

```
##        long      lat group order  region subregion
## 1 -87.46201 30.38968     1     1 alabama      <NA>
## 2 -87.48493 30.37249     1     2 alabama      <NA>
## 3 -87.52503 30.37249     1     3 alabama      <NA>
## 4 -87.53076 30.33239     1     4 alabama      <NA>
## 5 -87.57087 30.32665     1     5 alabama      <NA>
## 6 -87.58806 30.32665     1     6 alabama      <NA>
```

Plotting state murder rates

- Task: plot a map of the US states and fill in each state with a color representing the murder rate in that state using the `murders` dataset
- Need to use state map data and add the murder rate for each state to the map data frame

```
murders <- murders %>% mutate(region = tolower(state))
head(murders)
```

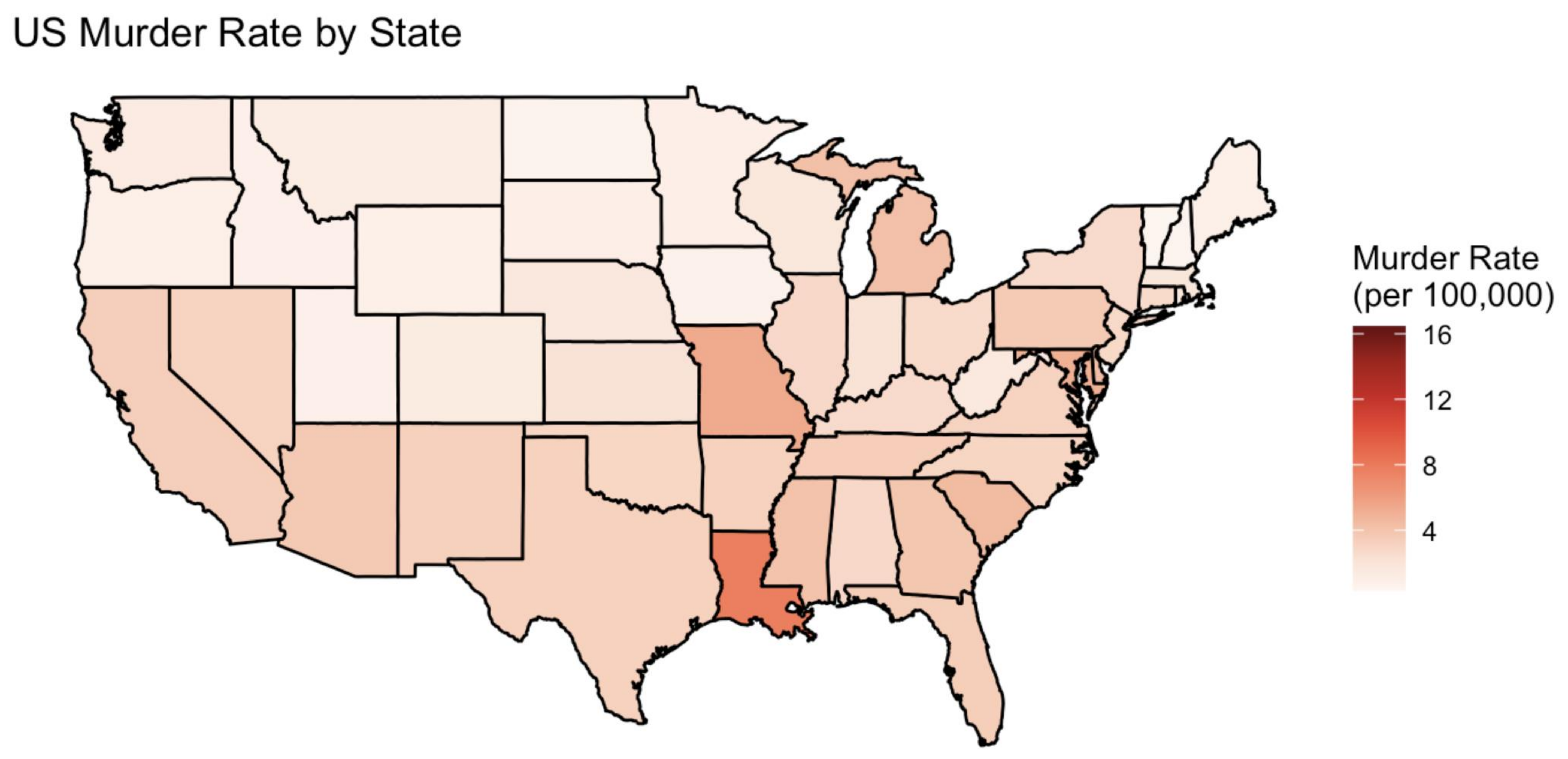
```
##        state abb     region population total     rate
## 1    Alabama  AL    alabama    4779736   135 2.824424
## 2     Alaska  AK     alaska     710231    19 2.675186
## 3    Arizona  AZ    arizona    6392017   232 3.629527
## 4   Arkansas  AR   arkansas    2915918    93 3.189390
## 5 California  CA california   37253956  1257 3.374138
## 6   Colorado  CO   colorado    5029196    65 1.292453
```

```
merged_data <- left_join(states_map, murders, by = "region")
head(merged_data)
```

```
##        long      lat group order  region subregion   state abb population total
## 1 -87.46201 30.38968     1     1 alabama      <NA> Alabama  AL    4779736   135
## 2 -87.48493 30.37249     1     2 alabama      <NA> Alabama  AL    4779736   135
## 3 -87.52503 30.37249     1     3 alabama      <NA> Alabama  AL    4779736   135
## 4 -87.53076 30.33239     1     4 alabama      <NA> Alabama  AL    4779736   135
## 5 -87.57087 30.32665     1     5 alabama      <NA> Alabama  AL    4779736   135
## 6 -87.58806 30.32665     1     6 alabama      <NA> Alabama  AL    4779736   135
##       rate
## 1 2.824424
## 2 2.824424
## 3 2.824424
## 4 2.824424
## 5 2.824424
## 6 2.824424
```

Plotting state murder rates

```r
merged_data %>%
ggplot(aes(long, lat, group = group, fill = rate)) +
  geom_polygon(color = "black") +
  scale_fill_gradientn(colors = brewer.pal(9, "Reds")) +
  theme_void() +
  labs(title = "US Murder Rate by State",
       fill = "Murder Rate \n(per 100,000)") +
  coord_fixed(1.3)
```



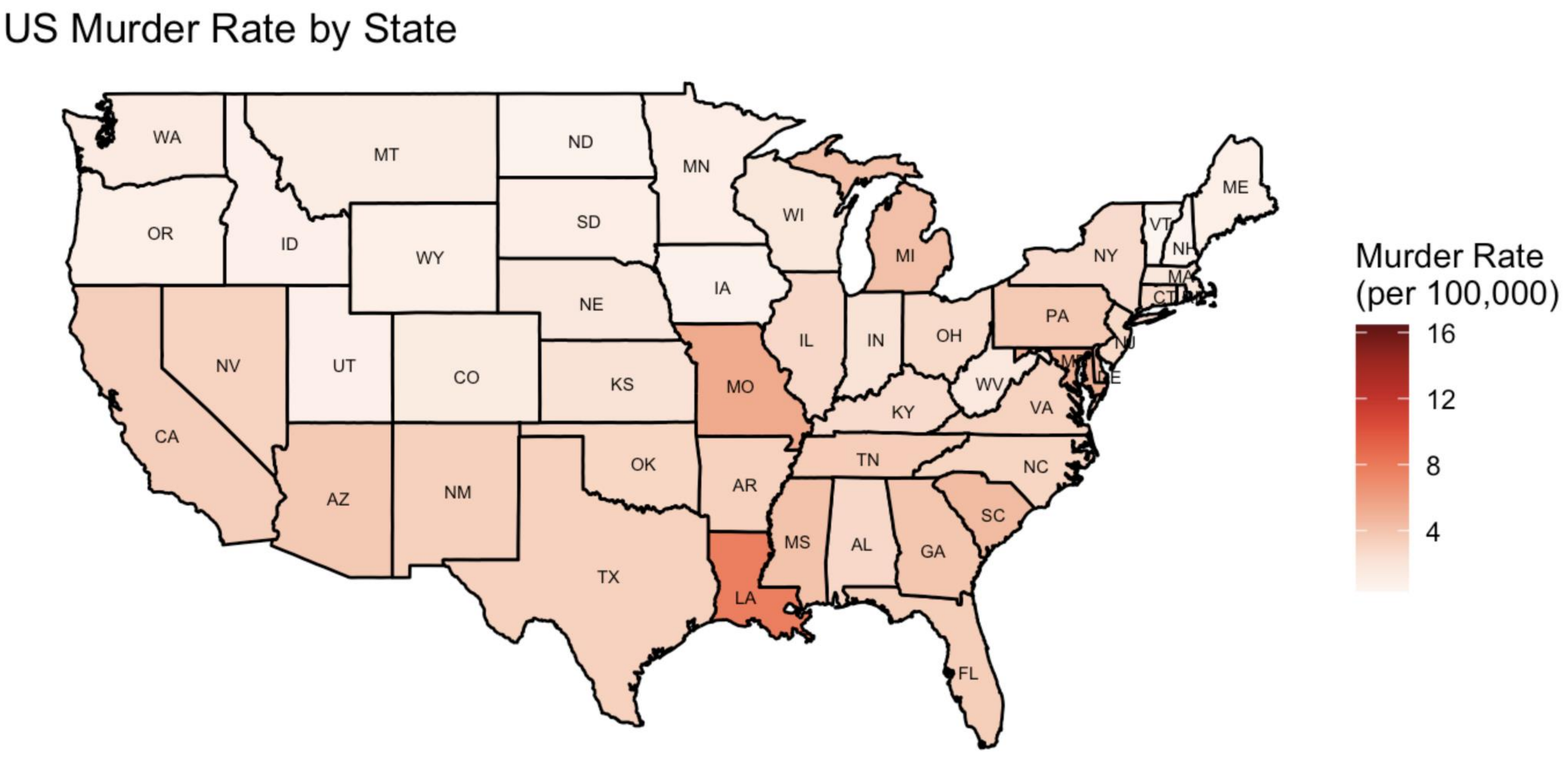
Adding state labels

```r
library(ggrepel)
centroids <- data.frame(region = tolower(state.name),
                        long = state.center$x,
                        lat = state.center$y)

centroids$abb <- state.abb[match(centroids$region,
                                 tolower(state.name))]

centroids <- centroids %>%
  filter(!region %in% c("alaska", "hawaii"))

merged_data %>%
ggplot(aes(long, lat, group = group, fill = rate)) +
  geom_polygon(color = "black") +
  scale_fill_gradientn(colors = brewer.pal(9, "Reds")) +
  theme_void() +
  labs(title = "US Murder Rate by State",
       fill = "Murder Rate \n(per 100,000)") +
  coord_fixed(1.3) +
  with(centroids,
       annotate(geom = "text", x = long, y = lat, label = abb,
                size = 2, color = "black", family = "Arial"))
```



Summary

- `map_data("states")` can be used to create a map of the contiguous United States
- The `tolower` function makes all letters in a string lowercase
- `state.center$x` and `state.center$y` contain the centoid longitude and latitude points for each US state
- Centroid locations can be used to plot state labels

Maps

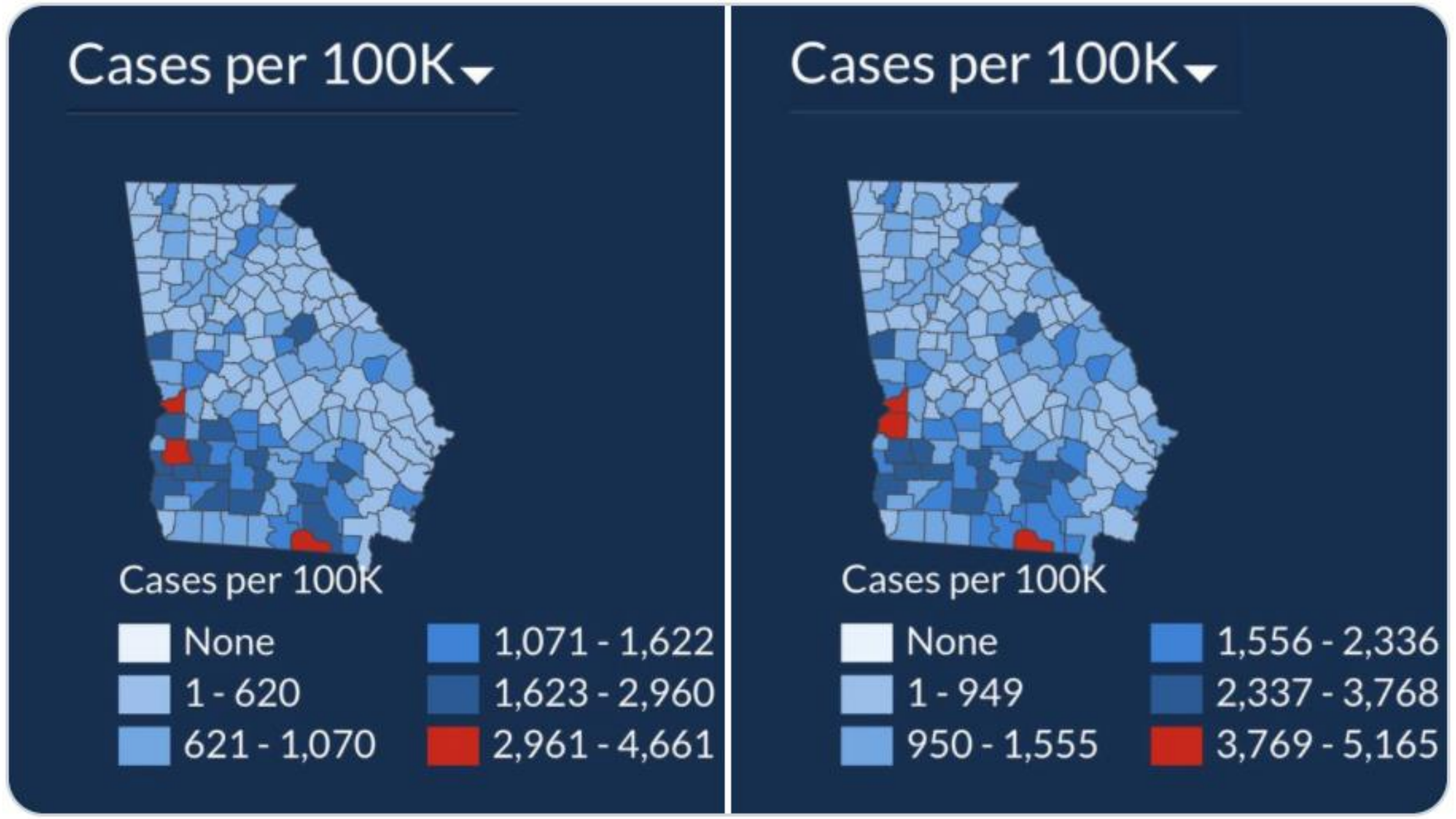
Case study: COVID-19

Heather Mattie, PhD

Misleading visualization

- In 2020 the Georgia Department of Public Health posted daily map visualizations of COVID-19 case counts in each county in Georgia
- On July 17, 2020, a tweet comparing the map on two different days pointed out an issue

In just 15 days the total number of #COVID19 cases in Georgia is up 49%, but you wouldn't know it from looking at the state's data visualization map of cases. The first map is July 2. The second is today. Do you see a 50% case increase? Can you spot how they're hiding it? 1/



5:24 PM · Jul 17, 2020 · Twitter for iPhone

Misleading visualization

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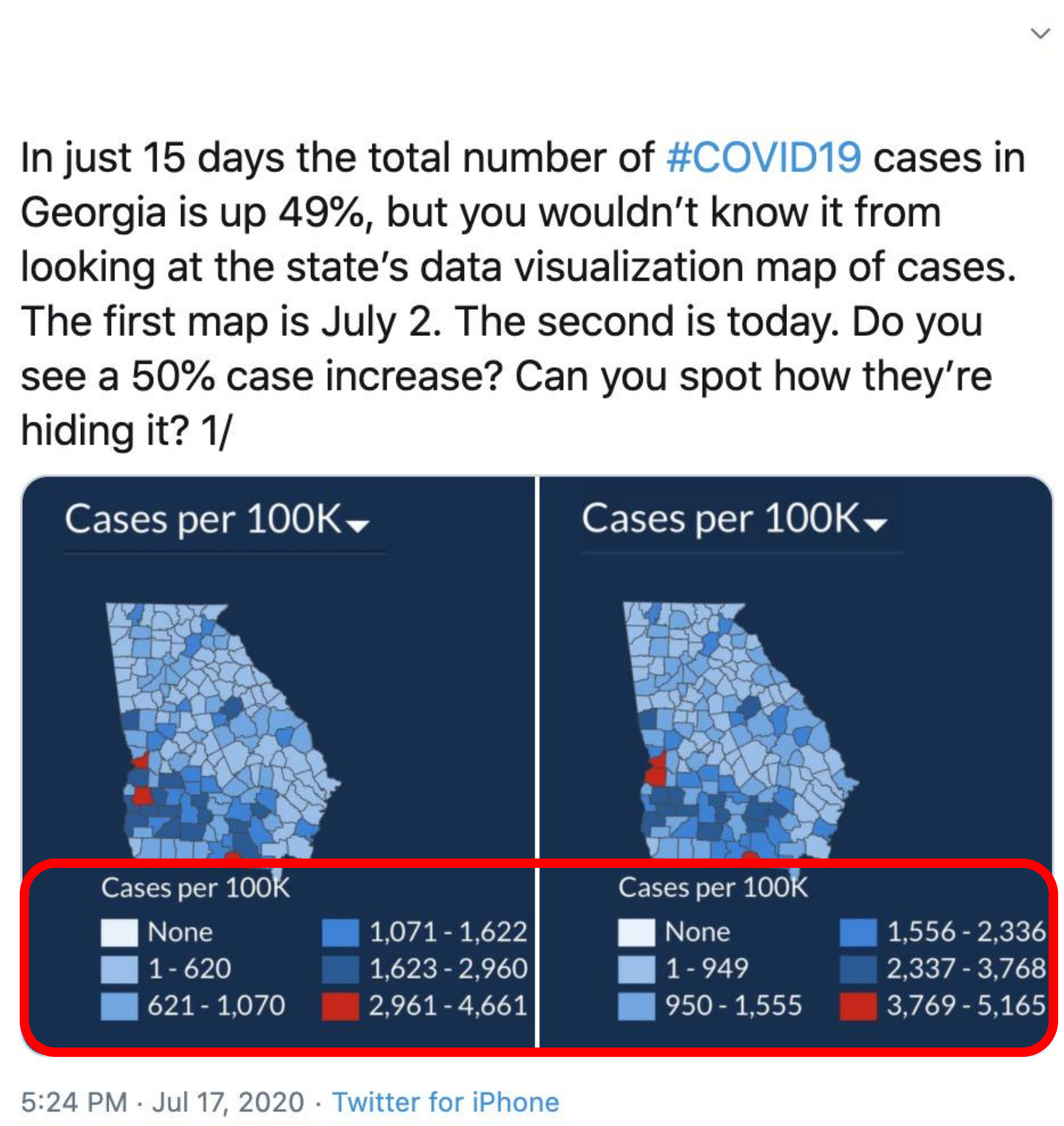


Data

We will need to combine three different datasets to recreate this plot

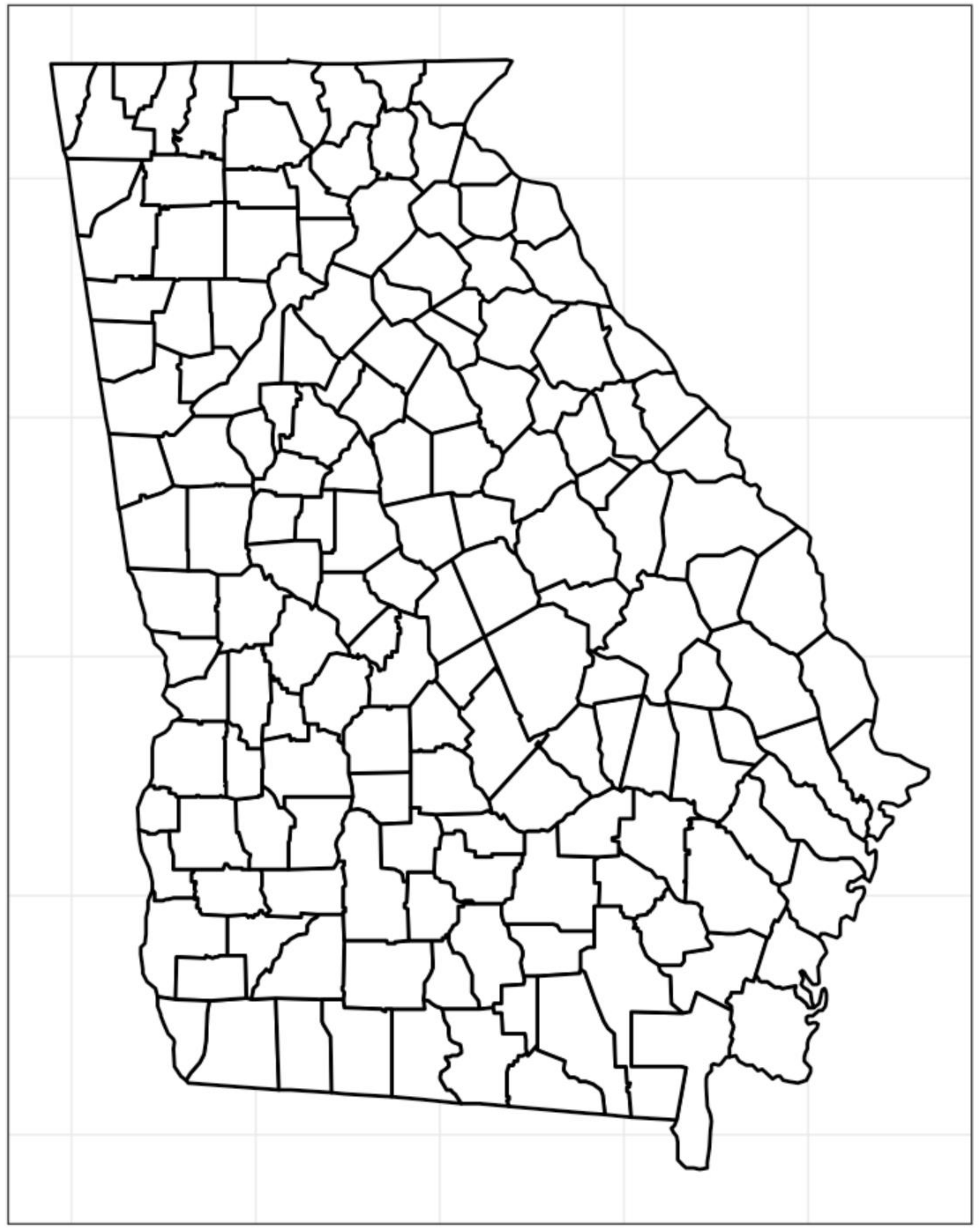
- Georgia county map data
  - From the `maps` package
- Daily COVID-19 case counts for each Georgia county
  - From New York Times COVID-19 data repository
- Population of each Georgia county
  - From the Georgia Department of Public Health (saved as csv)

Georgia county map data

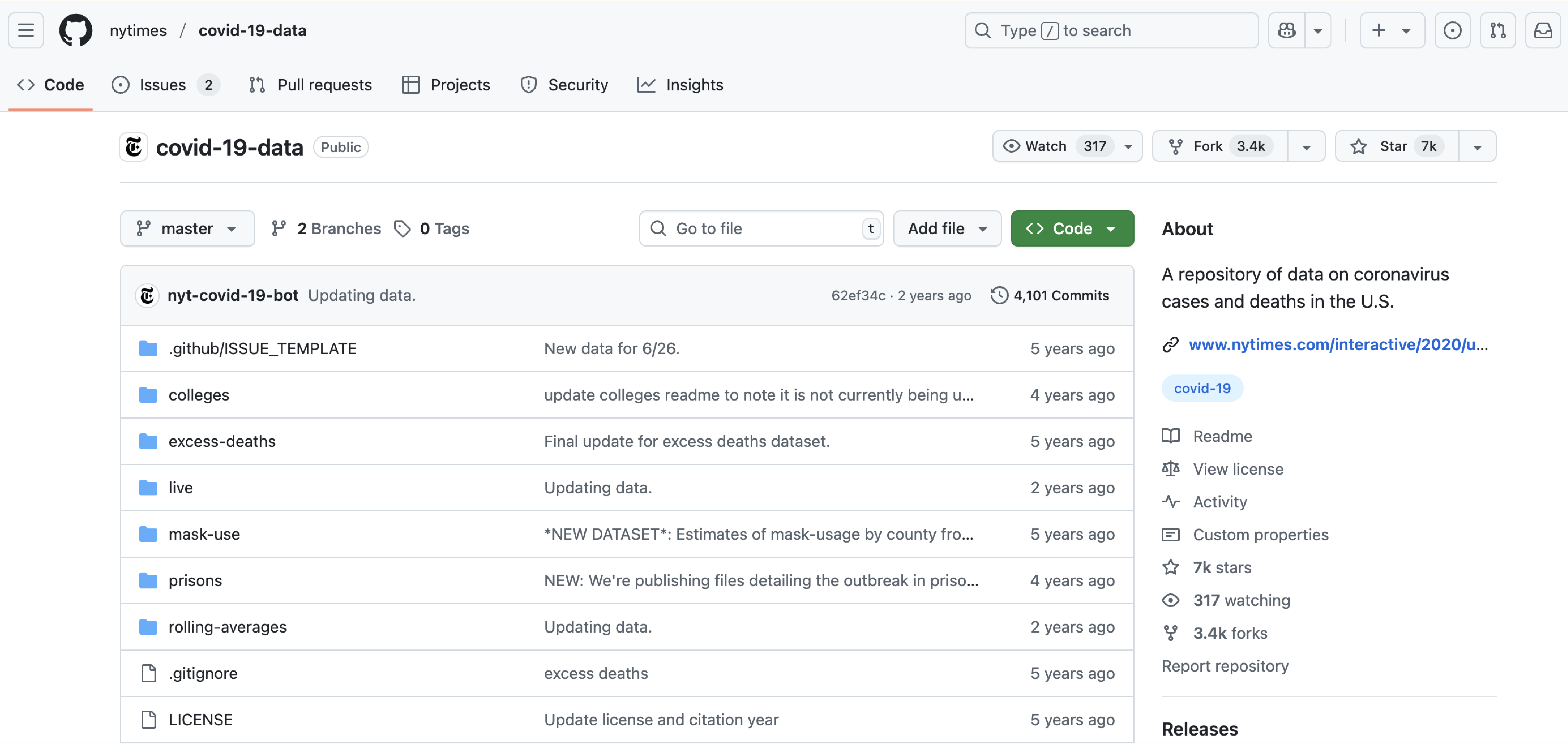
```r
GeorgiaCounty <- map_data("county", "georgia")
head(GeorgiaCounty)
```

```
##        long      lat group order  region subregion
## 1 -82.44862 31.94813     1     1 georgia   appling
## 2 -82.42570 31.94813     1     2 georgia   appling
## 3 -82.40852 31.94240     1     3 georgia   appling
## 4 -82.39706 31.94240     1     4 georgia   appling
## 5 -82.38560 31.93094     1     5 georgia   appling
## 6 -82.35122 31.91948     1     6 georgia   appling
```

```r
GeorgiaCounty %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "white", color = "black") +
  theme(panel.grid.major = element_blank(),
        panel.background = element_blank(),
        axis.title = element_blank(),
        axis.text = element_blank(),
        axis.ticks = element_blank()) +
  coord_fixed(1.3)
```



COVID-19 data



COVID-19 data

```r
url <- "https://raw.githubusercontent.com/nytimes/covid-19-data/master/us-counties.csv"
cases <- read_csv(url)
head(cases)
```

```
## # A tibble: 6 × 6
##   date       county    state      fips  cases deaths
##   <date>     <chr>     <chr>      <chr> <dbl>  <dbl>
## 1 2020-01-21 Snohomish Washington 53061     1      0
## 2 2020-01-22 Snohomish Washington 53061     1      0
## 3 2020-01-23 Snohomish Washington 53061     1      0
## 4 2020-01-24 Cook      Illinois   17031     1      0
## 5 2020-01-24 Snohomish Washington 53061     1      0
## 6 2020-01-25 Orange    California 06059     1      0
```

COVID-19 data

```
dates <- c(ymd("2020-07-02", "2020-07-17"))
georgia_cases <- cases %>% filter(state == "Georgia", date %in% dates)
head(georgia_cases)
```

```
## # A tibble: 6 × 6
##   date       county    state   fips  cases deaths
##   <date>     <chr>     <chr>   <chr> <dbl>  <dbl>
## 1 2020-07-02 Appling   Georgia 13001   266     14
## 2 2020-07-02 Atkinson  Georgia 13003   155      2
## 3 2020-07-02 Bacon     Georgia 13005   254      4
## 4 2020-07-02 Baker     Georgia 13007    43      3
## 5 2020-07-02 Baldwin   Georgia 13009   543     34
## 6 2020-07-02 Banks     Georgia 13011   140      1
```

Georgia county population data

```r
pop_data <- read_csv("county_pop.csv")
head(pop_data)
```

```
## # A tibble: 6 × 12
##   county_name cases county_id `State FIPS code` `County FIPS code` population
##   <chr>       <dbl> <chr>     <chr>             <chr>                   <dbl>
## 1 Appling      1054 US-13001  13                1                       18561
## 2 Atkinson      447 US-13003  13                3                        8330
## 3 Bacon         602 US-13005  13                5                       11404
## 4 Baker          84 US-13007  13                7                        3116
## 5 Baldwin      2132 US-13009  13                9                       44428
## 6 Banks         501 US-13011  13                11                      19982
## # ℹ 6 more variables: hospitalization <dbl>, deaths <dbl>, `case rate` <dbl>,
## #   `death rate` <dbl>, `14 day case rate` <dbl>, `14 day cases` <dbl>
```

Joining data frames

1. Join the county population data to the COVID-19 cases data

```
pop_data <- pop_data %>% select(county_name, population)

data_full <- left_join(georgia_cases, pop_data, by = c("county" = "county_name"))
head(data_full)
```

```
## # A tibble: 6 × 7
##   date       county   state   fips  cases deaths population
##   <date>     <chr>    <chr>   <chr> <dbl>  <dbl>      <dbl>
## 1 2020-07-02 Appling  Georgia 13001   266     14      18561
## 2 2020-07-02 Atkinson Georgia 13003   155      2       8330
## 3 2020-07-02 Bacon    Georgia 13005   254      4      11404
## 4 2020-07-02 Baker    Georgia 13007    43      3       3116
## 5 2020-07-02 Baldwin  Georgia 13009   543     34      44428
## 6 2020-07-02 Banks    Georgia 13011   140      1      19982
```

Joining data frames

1. Join the county population data to the COVID-19 cases data
2. Calculate cases per 100,000 population

```
data_full <- data_full %>% mutate(rate = 100000*(cases/population))
head(data_full)
```

```
## # A tibble: 6 × 8
##   date       county    state   fips  cases deaths population  rate
##   <date>     <chr>     <chr>   <chr> <dbl>  <dbl>      <dbl> <dbl>
## 1 2020-07-02 Appling   Georgia 13001   266     14      18561 1433.
## 2 2020-07-02 Atkinson  Georgia 13003   155      2       8330 1861.
## 3 2020-07-02 Bacon     Georgia 13005   254      4      11404 2227.
## 4 2020-07-02 Baker     Georgia 13007    43      3       3116 1380.
## 5 2020-07-02 Baldwin   Georgia 13009   543     34      44428 1222.
## 6 2020-07-02 Banks     Georgia 13011   140      1      19982  701.
```

Joining data frames

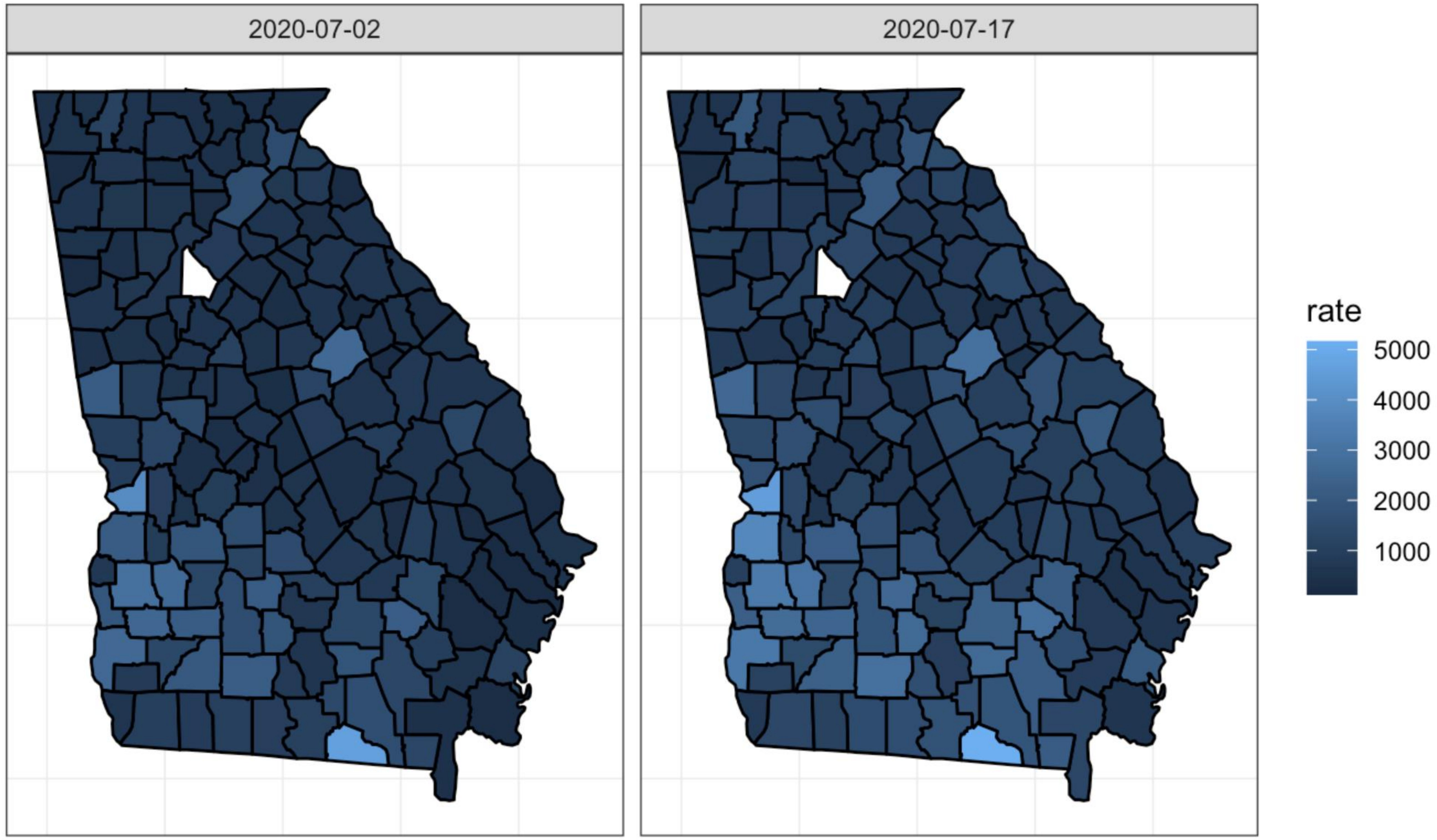
1. Join the county population data to the COVID-19 cases data
2. Calculate cases per 100,000 population
3. Join to map data

```
georgia_map <- data_full %>% mutate(county = str_to_lower(county)) %>%
  left_join(GeorgiaCounty, by = c("county" = "subregion"))

head(georgia_map)
```

```
## # A tibble: 6 × 13
##   date       county  state fips  cases deaths population  rate  long   lat group
##   <date>     <chr>   <chr> <chr> <dbl>  <dbl>      <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 2020-07-02 appling Geor… 13001   266     14      18561 1433. -82.4  31.9     1
## 2 2020-07-02 appling Geor… 13001   266     14      18561 1433. -82.4  31.9     1
## 3 2020-07-02 appling Geor… 13001   266     14      18561 1433. -82.4  31.9     1
## 4 2020-07-02 appling Geor… 13001   266     14      18561 1433. -82.4  31.9     1
## 5 2020-07-02 appling Geor… 13001   266     14      18561 1433. -82.4  31.9     1
## 6 2020-07-02 appling Geor… 13001   266     14      18561 1433. -82.4  31.9     1
## # ℹ 2 more variables: order <int>, region <chr>
```

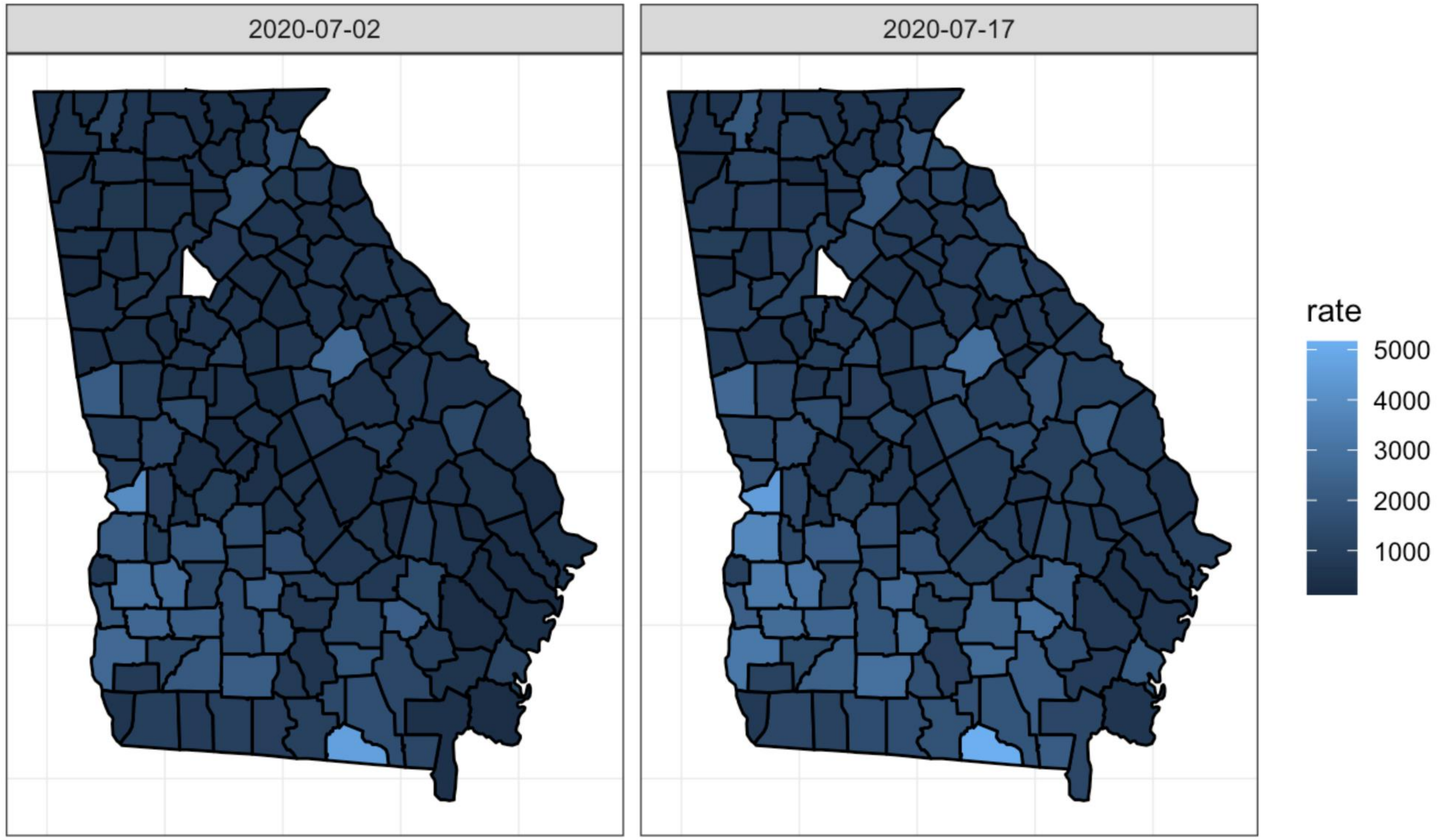
Data inconsistencies

```r
georgia_map %>%
  ggplot(aes(x = long, y = lat, group = group, fill = rate)) +
  geom_polygon(color = "black") +
  theme(panel.grid.major = element_blank(),
        panel.background = element_blank(),
        axis.title = element_blank(),
        axis.text = element_blank(),
        axis.ticks = element_blank()) +
  coord_fixed(1.3) +
  facet_grid( ~ date)
```



Data inconsistencies

```r
georgia_map %>%
  ggplot(aes(x = long, y = lat, group = group, fill = rate)) +
  geom_polygon(color = "black") +
  theme(panel.grid.major = element_blank(),
        panel.background = element_blank(),
        axis.title = element_blank(),
        axis.text = element_blank(),
        axis.ticks = element_blank()) +
  coord_fixed(1.3) +
  facet_grid( ~ date)
```



Problem: De Kalb county

Saved as "**de kalb**" in the **`GeorgiaCounty`** data frame, but as "**DeKalb**" in the in the **`georgia_cases`** data frame

Data inconsistencies

First solution attempt:

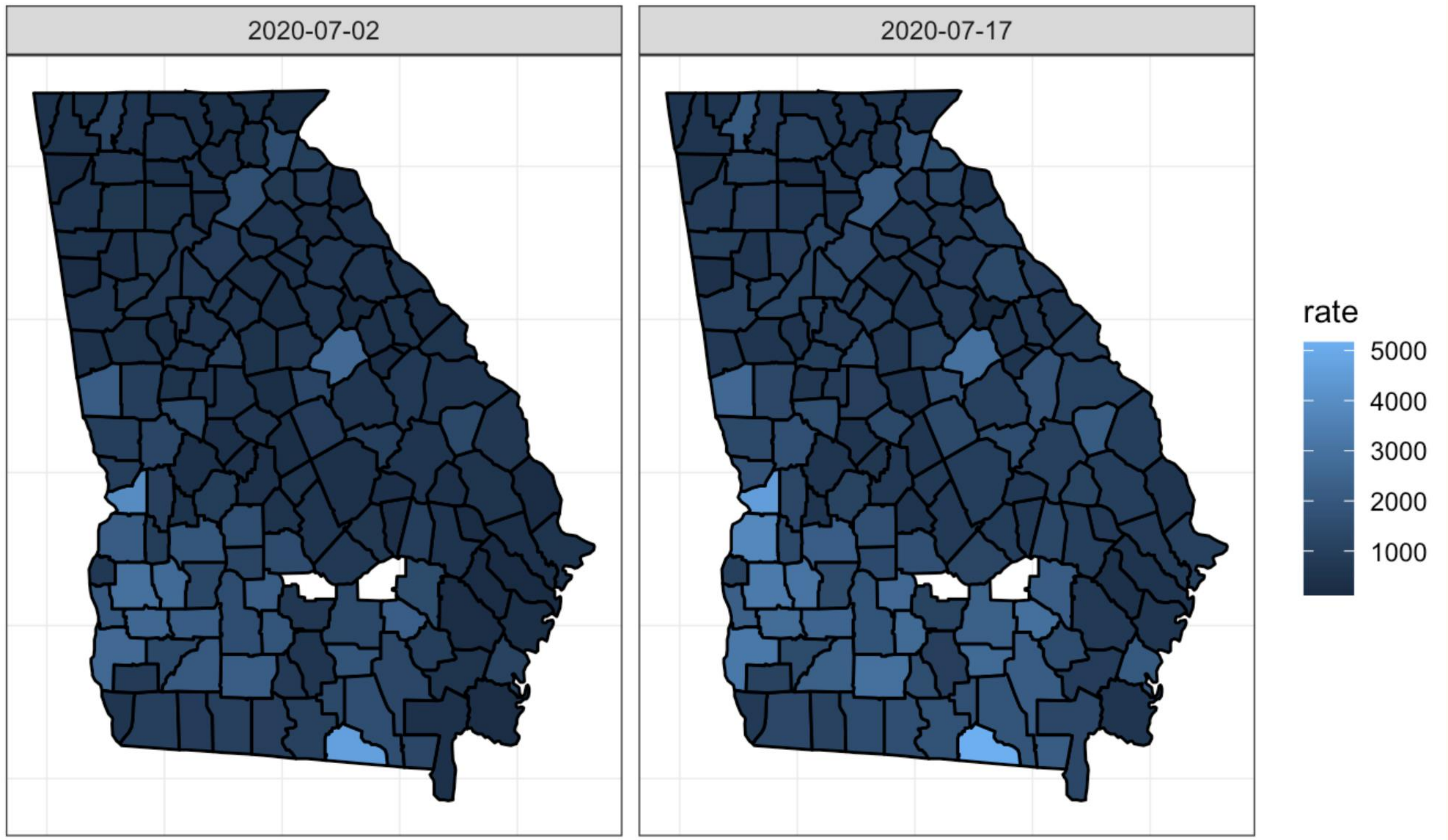
1. Remove all spaces in county names using the **`str_replace`** function
2. Convert all county names to lowercase using the **`str_to_lower`** function

```
GeorgiaCounty <- GeorgiaCounty %>%
  mutate(subregion = str_replace(subregion, " ", ""))

georgia_map <- data_full %>%
  mutate(county = str_to_lower(county)) %>%
  left_join(GeorgiaCounty, by = c("county" = "subregion"))

head(georgia_map)
```

```
## # A tibble: 6 × 13
##   date       county  state fips  cases deaths population  rate  long   lat group
##   <date>     <chr>   <chr> <chr> <dbl>  <dbl>      <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 2020-07-02 appling Geor… 13001   266     14      18561 1433. -82.4  31.9     1
## 2 2020-07-02 appling Geor… 13001   266     14      18561 1433. -82.4  31.9     1
## 3 2020-07-02 appling Geor… 13001   266     14      18561 1433. -82.4  31.9     1
## 4 2020-07-02 appling Geor… 13001   266     14      18561 1433. -82.4  31.9     1
## 5 2020-07-02 appling Geor… 13001   266     14      18561 1433. -82.4  31.9     1
## 6 2020-07-02 appling Geor… 13001   266     14      18561 1433. -82.4  31.9     1
## # ℹ 2 more variables: order <int>, region <chr>
```

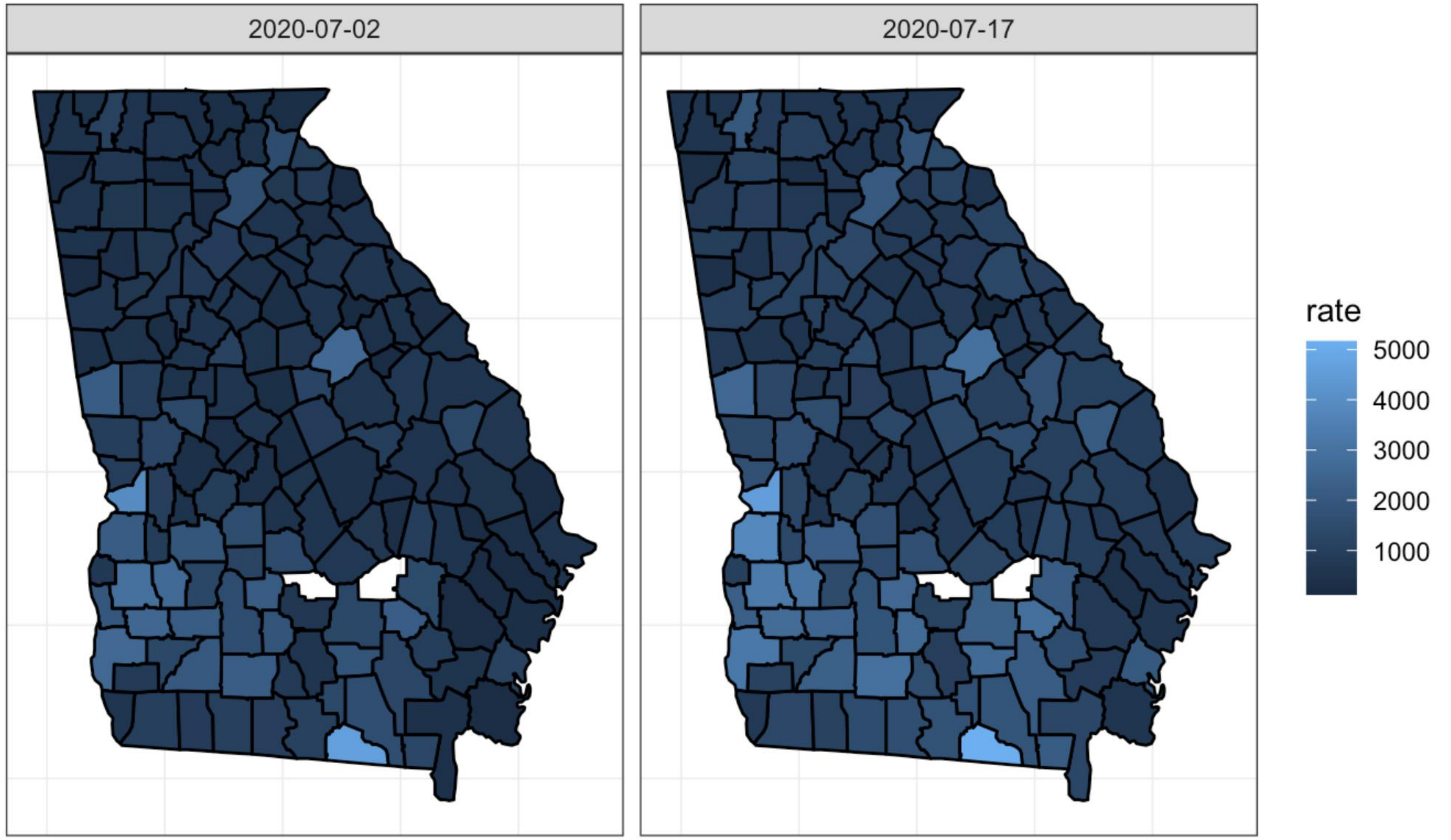
Data inconsistencies

```r
georgia_map %>%
  ggplot(aes(x = long, y = lat, group = group, fill = rate)) +
  geom_polygon(color = "black") +
  theme(panel.grid.major = element_blank(),
        panel.background = element_blank(),
        axis.title = element_blank(),
        axis.text = element_blank(),
        axis.ticks = element_blank()) +
  coord_fixed(1.3) +
  facet_grid( ~ date)
```



Data inconsistencies

```r
georgia_map %>%
  ggplot(aes(x = long, y = lat, group = group, fill = rate)) +
  geom_polygon(color = "black") +
  theme(panel.grid.major = element_blank(),
        panel.background = element_blank(),
        axis.title = element_blank(),
        axis.text = element_blank(),
        axis.ticks = element_blank()) +
  coord_fixed(1.3) +
  facet_grid( ~ date)
```



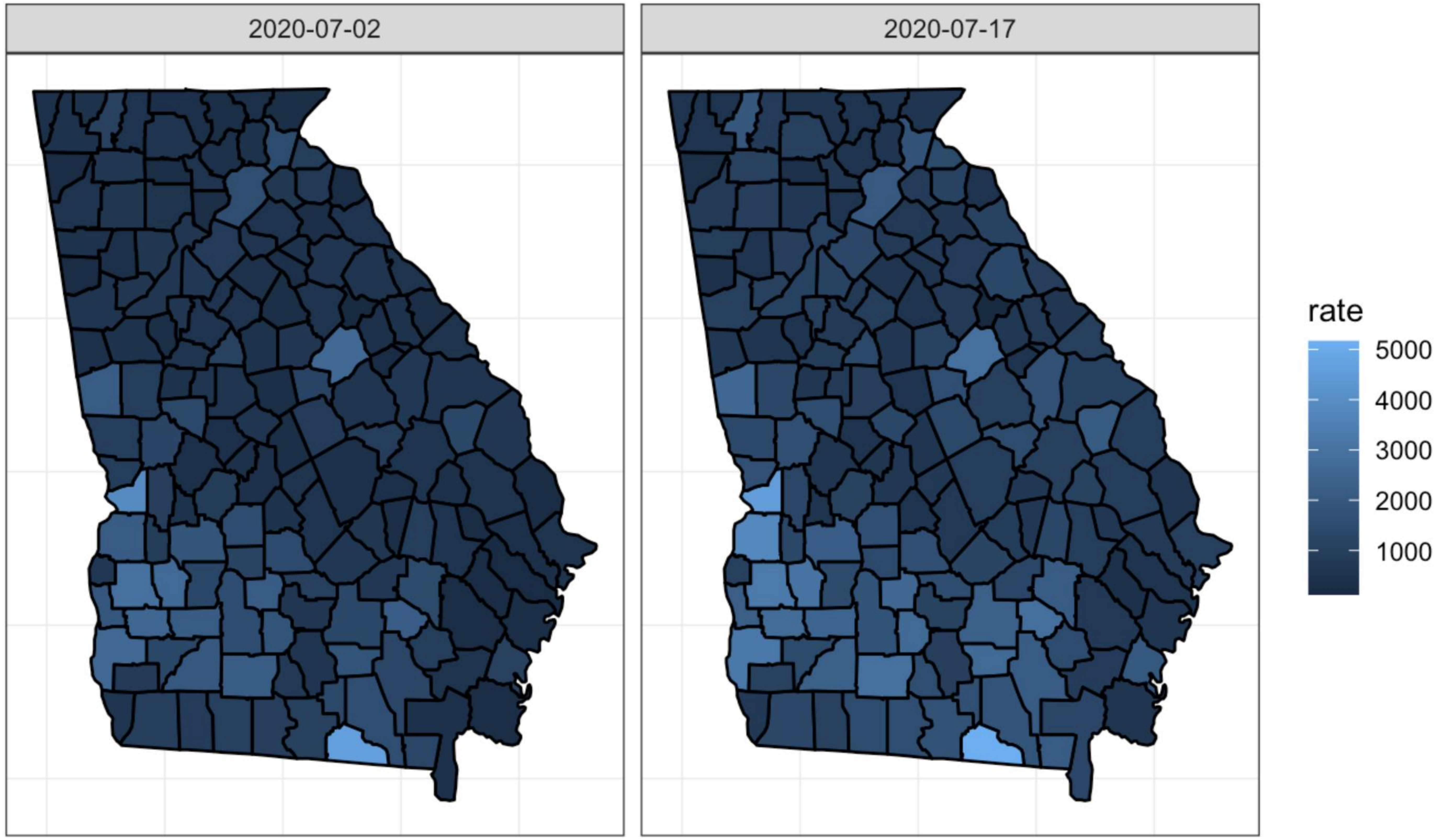
Problem: **Ben Hill** and **Jeff Davis** counties

Both data frames had a space between the two names but the space was removed for `GeorgiaCounty` and not for `georgia_cases`

Data inconsistencies

```r
data_full <- data_full %>%
  mutate(county = str_replace(county, " ", ""))

georgia_map <- data_full %>%
  mutate(county = str_to_lower(county)) %>%
  left_join(GeorgiaCounty, by = c("county" = "subregion"))
```

```r
georgia_map %>%
  ggplot(aes(x = long, y = lat, group = group, fill = rate)) +
  geom_polygon(color = "black") +
  theme(panel.grid.major = element_blank(),
        panel.background = element_blank(),
        axis.title = element_blank(),
        axis.text = element_blank(),
        axis.ticks = element_blank()) +
   coord_fixed(1.3) +
   facet_grid( ~ date)
```



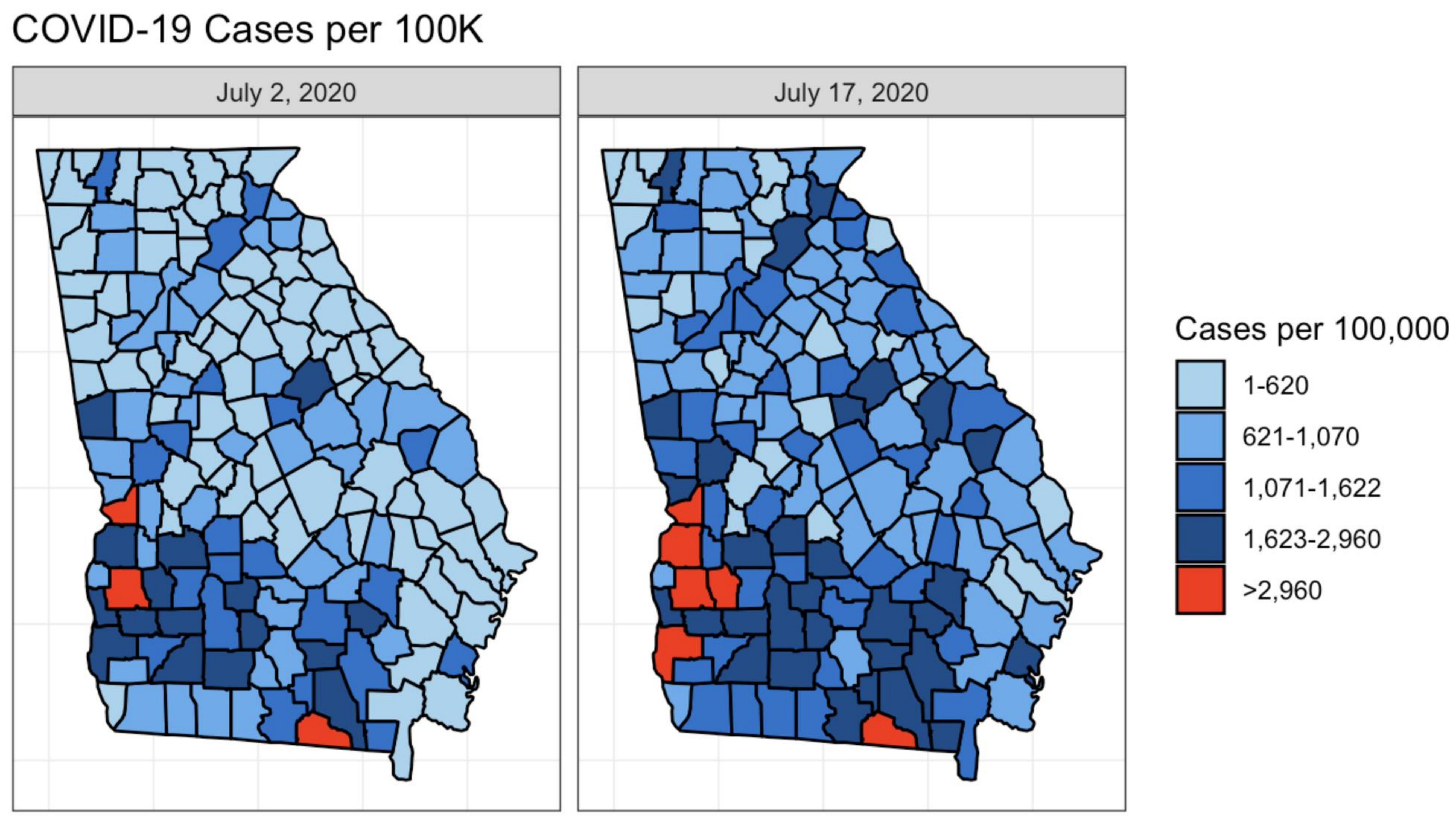
Customizing legends

```
georgia_map <- georgia_map %>%
  mutate(manual_fill = cut(rate, breaks = c(0, 620, 1070, 1622, 2960, Inf),
                           labels = c("1-620", "621-1,070", "1,071-1,622",
                                      "1,623-2,960", ">2,960"),
                           right = TRUE))
head(georgia_map)
```

```
## # A tibble: 6 × 14
##   date       county  state fips  cases deaths population  rate  long   lat group
##   <date>     <chr>   <chr> <chr> <dbl>  <dbl>      <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 2020-07-02 appling Geor… 13001   266     14      18561 1433. -82.4  31.9     1
## 2 2020-07-02 appling Geor… 13001   266     14      18561 1433. -82.4  31.9     1
## 3 2020-07-02 appling Geor… 13001   266     14      18561 1433. -82.4  31.9     1
## 4 2020-07-02 appling Geor… 13001   266     14      18561 1433. -82.4  31.9     1
## 5 2020-07-02 appling Geor… 13001   266     14      18561 1433. -82.4  31.9     1
## 6 2020-07-02 appling Geor… 13001   266     14      18561 1433. -82.4  31.9     1
## # ℹ 3 more variables: order <int>, region <chr>, manual_fill <fct>
```

Customizing color palettes

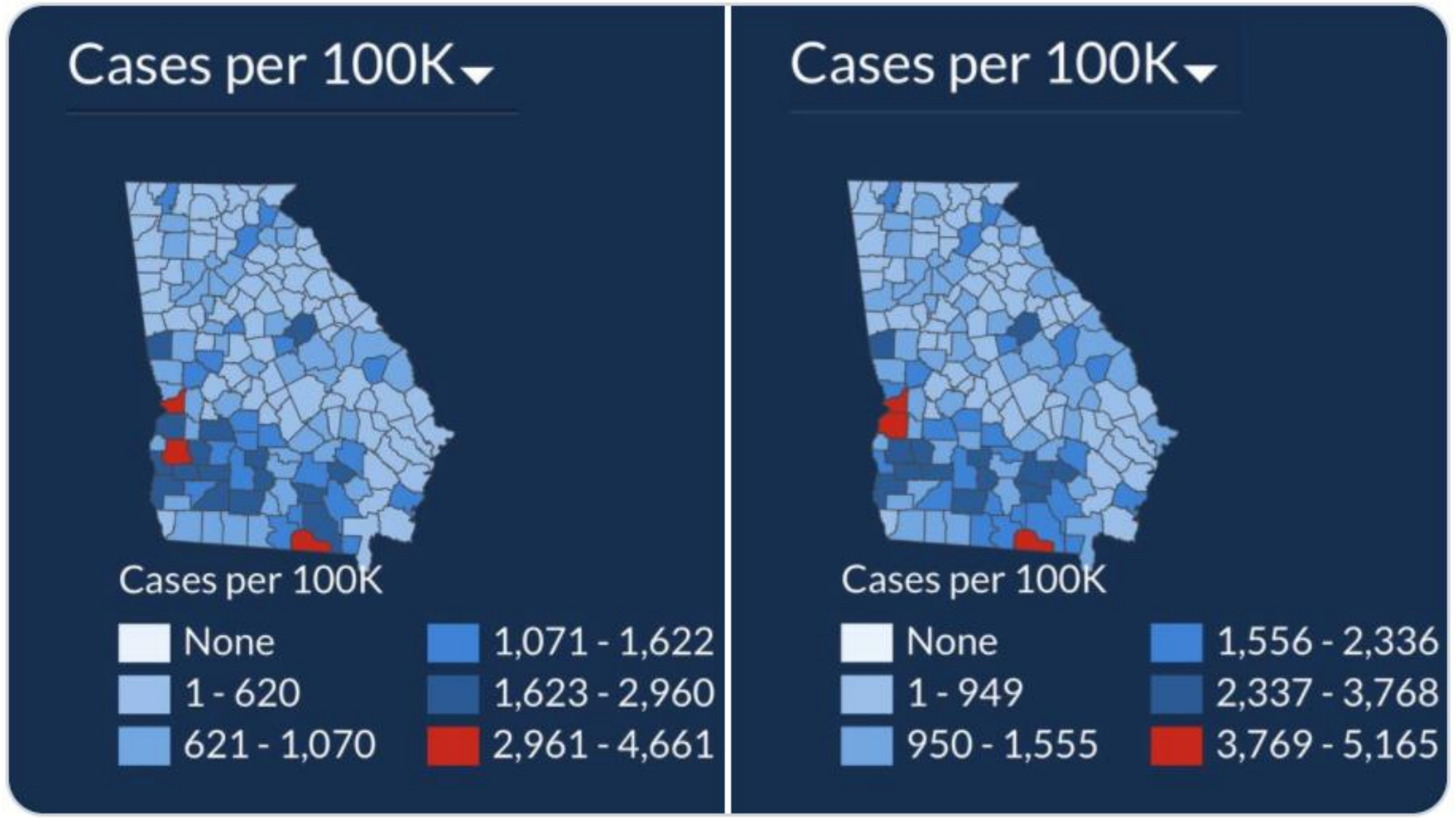
```r
pal <- c("lightskyblue2", "steelblue2", "dodgerblue3", "dodgerblue4", "red")

date.labs <- c("July 2, 2020", "July 17, 2020")
names(date.labs) <- c("2020-07-02", "2020-07-17")

georgia_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(aes(fill = manual_fill), color = "black")  +
  scale_fill_manual(name = "Cases per 100,000", values = pal) +
  coord_fixed(1.3) +
  theme(panel.grid.major = element_blank(),
        panel.background = element_blank(),
        axis.title = element_blank(),
        axis.text = element_blank(),
        axis.ticks = element_blank()) +
  ggtitle("COVID-19 Cases per 100K") +
  facet_grid(. ~ date, labeller = labeller(date = date.labs))
```



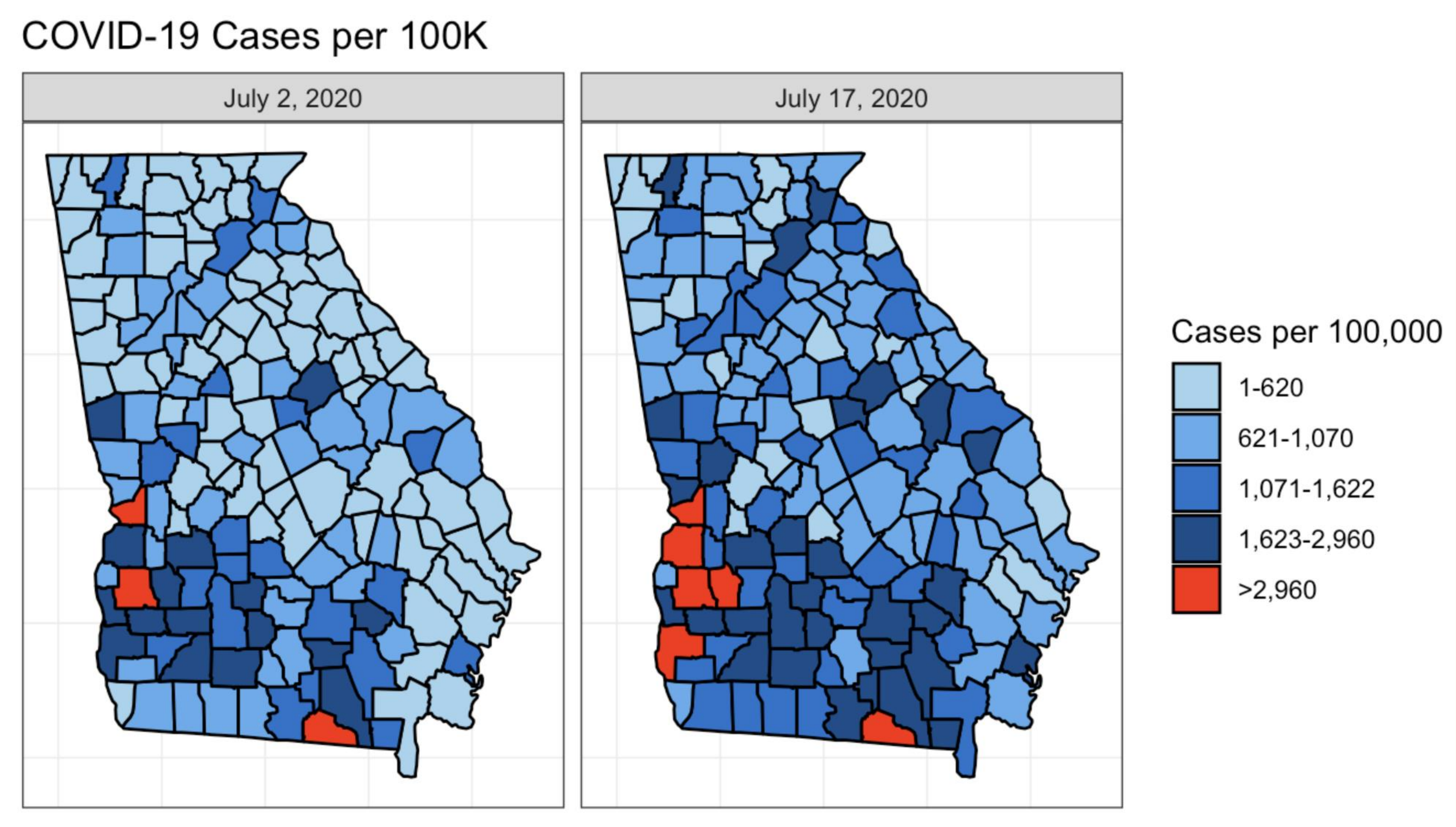
Plot comparison

Georgia Person
@andishehnouraee

In just 15 days the total number of #COVID19 cases in Georgia is up 49%, but you wouldn't know it from looking at the state's data visualization map of cases. The first map is July 2. The second is today. Do you see a 50% case increase? Can you spot how they're hiding it? 1/



5:24 PM · Jul 17, 2020 · Twitter for iPhone



Summary

- Plots of maps can be misleading, especially when plotting changes over time
- There should only be one legend per plot if subplots are related
- Color palettes can be customized